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Safety data sheet

BASF safety data sheet. This document has been drafted following generic rules for safety data sheets. It does not replace the safety data sheet provided according to Regulation (EC) No 1907/2006.

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Product: **Basoflux® RD 5119 T**

(ID no. 30686023/SDS_GEN_EU/EN)
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SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Basoflux® RD 5119 T

UFI: 01KH-HF7V-D00R-01FC

1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Performance Chemicals for Oilfield Applications

For the detailed identified uses of the product see appendix of the safety data sheet.

1.3. Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Global Oilfield Chemicals

Telephone: +49 621 60-0
E-mail address: RegXcellenceOilfieldChemicals@basf.com

1.4. Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

SECTION 2: Hazards Identification

2.1. Classification of the substance or mixture

For the classification of the mixture the following methods have been applied: extrapolation on the concentration levels of the hazardous substances, on basis of test results and after evaluation of experts. The methodologies used are mentioned at the respective test results.

According to Regulation (EC) No 1272/2008 [CLP]

STOT SE 3	H336 May cause drowsiness or dizziness.
Aquatic Chronic 3	H412 Harmful to aquatic life with long lasting effects.

For the classifications not written out in full in this section the full text can be found in section 16.

2.2. Label elements

According to Regulation (EC) No 1272/2008 [CLP]

Pictogram:

Signal Word:

Warning

Hazard Statement:

H336	May cause drowsiness or dizziness.
H412	Harmful to aquatic life with long lasting effects.

Precautionary Statements (Prevention):

P271	Use only outdoors or in a well-ventilated area.
P273	Avoid release to the environment.
P261	Avoid breathing mist or vapour or spray.

Precautionary Statements (Response):

P312	Call a POISON CENTER or physician if you feel unwell.

Precautionary Statements (Storage):

P403 + P233	Store in a well-ventilated place. Keep container tightly closed.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

Labeling of special preparations (GHS):

EUH066: Repeated exposure may cause skin dryness or cracking.
Product contains the following components and may cause carcinogenicity: Naphthalene

Hazard determining component(s) for labelling: Hydrocarbons, C11-C14, n-alkanes, isoalkanes, cyclics, <2% aromatics, Hydrocarbons, C10, aromatics, <1% naphthalene, naphthalene (as a component of hydrocarbons, C10, aromatics), 1,2,4-trimethylbenzene

2.3. Other hazards

According to Regulation (EC) No 1272/2008 [CLP]

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.
The product does not contain a substance above legal limits fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.
Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.

SECTION 3: Composition/Information on Ingredients

3.1. Substances

Not applicable

3.2. Mixtures

Chemical nature

Preparation based on:polyacrylate, solvent(s)

Regulatory relevant ingredients

Hydrocarbons, C11-C14, n-alkanes, isoalkanes, cyclics, <2% aromatics

Content (W/W): >= 25 % - < 50 %	Asp. Tox. 1
REACH registration number: 01-2119456620-43	H304
	EUH066

Hydrocarbons, C10, aromatics, <1% naphthalene

Content (W/W): >= 20 % - < 25 %	Asp. Tox. 1
EC-Number: 918-811-1	STOT SE 3 (drowsiness and dizziness)
REACH registration number: 01-2119463583-34	Aquatic Chronic 2
	H411, H304, H336
	EUH066

1,2,4-trimethylbenzene

Content (W/W): >= 1 % - < 3 %	Asp. Tox. 1
CAS Number: 95-63-6	Flam. Liq. 3
EC-Number: 202-436-9	Acute Tox. 4 (Inhalation - vapour)
REACH registration number: 01-211-9472135-42	Skin Irrit. 2
	Eye Irrit. 2
INDEX-Number: 601-043-00-3	Aquatic Chronic 2
	STOT SE 3 (irr. to respiratory syst.)
	H226, H319, H315, H332, H304, H335, H411

naphthalene (as a component of hydrocarbons, C10, aromatics)

Content (W/W): > 0 % - < 0.3 %	Flam. Sol. 2
CAS Number: 91-20-3	Acute Tox. 4 (oral)
EC-Number: 202-049-5	Carc. 2
INDEX-Number: 601-052-00-2	Aquatic Acute 1
	Aquatic Chronic 1
Substance with EU occupational exposure limit	M-factor acute: 1
	M-factor chronic: 1
	H228, H302, H351, H400, H410

For the classifications not written out in full in this section, including the hazard classes and the hazard statements, the full text is listed in section 16.

SECTION 4: First-Aid Measures

4.1. Description of first aid measures

Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

4.2. Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

4.3. Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

SECTION 5: Fire-Fighting Measures

5.1. Extinguishing media

Suitable extinguishing media:
dry powder, foam

5.2. Special hazards arising from the substance or mixture

Endangering substances: harmful vapours, carbon oxides, hydrocarbons
Advice: Generation of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

5.3. Advice for fire-fighters

Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:
The degree of risk is governed by the burning substance and the fire conditions. Contaminated extinguishing water must be disposed of in accordance with official regulations.

SECTION 6: Accidental Release Measures

6.1. Personal precautions, protective equipment and emergency procedures

6.2. Environmental precautions

Contain contaminated water/firefighting water. Do not discharge into drains/surface waters/groundwater.

6.3. Methods and material for containment and cleaning up

For large amounts: Pump off product.
For residues: Pick up with suitable absorbent material. Dispose of absorbed material in accordance with regulations.

6.4. Reference to other sections

Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

SECTION 7: Handling and Storage

7.1. Precautions for safe handling

Ensure thorough ventilation of stores and work areas. Avoid skin contact. Avoid inhalation of the product. Wear suitable personal protective equipment.

No eating, drinking, smoking or tobacco use at the place of work. Remove contaminated clothing and protective equipment before entering eating areas. Wash hands before breaks and at end of work.

Protection against fire and explosion:
Sources of ignition should be kept well clear. Take precautionary measures against static discharges. If delivered in plastic packing, highest permissable emptying temperature is 5 Kelvin below the flash point.

7.2. Conditions for safe storage, including any incompatibilities

Segregate from foods and animal feeds. Segregate from nitrites.

Suitable materials for containers: Stove-lacquer RDL 50, Stove-lacquer R 78433, Stainless steel 1.4311, Stainless steel 1.4361, Stainless steel 1.4401, Stainless steel 1.4404, glass, Stainless steel 1.4301 (V2), Stainless steel 1.4306 (V2A), Stainless steel 1.4307
Further information on storage conditions: Keep container tightly closed and in a cool place.

Storage stability:
Storage temperature: <= 50 °C

The packed product is not damaged by low temperatures or by frost.
Protect from temperatures above:50 °C
Changes in the properties of the product may occur if substance/product is stored above indicated temperature for extended periods of time.

7.3. Specific end use(s)

See lead substance(s) exposure scenarios attached as annex to the safety data sheet.

SECTION 8: Exposure Controls/Personal Protection

8.1. Control parameters

Components with occupational exposure limits

91-20-3: Naphthalene
95-63-6: 1,2,4-trimethylbenzene
64742-47-8: Distillates (petroleum), hydro- treated light
64742-94-5: Solvent naphtha (petroleum), heavy arom.
Hydrocarbons, C11-C14, n-alkanes, isoalkanes, cyclics, <2% aromatics

PNEC

Data refer to the lead substance

Components with PNEC

Hydrocarbons, C10, aromatics, <1% naphthalene
No PNEC value available.

DNEL

Data refer to the lead substance

Components with DNEL

Hydrocarbons, C10, aromatics, <1% naphthalene
consumer: Long-term exposure- systemic effects, dermal: 7.5 mg/kg
consumer: Long-term exposure- systemic effects, oral: 7.5 mg/kg
consumer: Long-term exposure- systemic effects, Inhalation: 32 mg/m3

worker: Long-term exposure- systemic effects, dermal: 12.5 mg/kg
worker: Long-term exposure- systemic effects, Inhalation: 151 mg/m3

8.2. Exposure controls

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for lower concentrations or short-term effect: Gas filter for gases/vapours of organic compounds (boiling point >65 °C, e. g. EN 14387 Type A)

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures

Handle in accordance with good industrial hygiene and safety practice.

Environmental exposure controls

Do not discharge product into the environment without control.

SECTION 9: Physical and Chemical Properties

9.1. Information on basic physical and chemical properties

State of matter:	liquid	
Form:	liquid to waxy	
Colour:	colourless to yellowish	
Odour:	mineral oil-like	
Odour threshold:	not determined	
Melting point:	17 - 23 °C	
Boiling point:	> 190 °C	
Flammability:	Combustible liquid.	(derived from flash point)
Lower explosion limit:	For liquids not relevant for classification and labelling., The lower explosion point may be 5 - 15 °C below the flash point.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	

Flash point:	62.5 °C The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.	(ISO 2719)
Auto-ignition temperature:	> 230 °C	(DIN 51794)
Thermal decomposition:	375 °C, > 200 J/g (DSC (DIN 51007)) self-accelerating reaction	
pH value:	not applicable, insoluble	
Viscosity, kinematic:	approx. 1,060 mm2/s (23 °C)	(calculated (from dynamic viscosity))
Viscosity, dynamic:	932 mPa.s (23 °C)	
Solubility in water:	dispersible when hot	
Solubility (qualitative) solvent(s):	aromatic hydrocarbons soluble	
Partitioning coefficient n-octanol/water (log Kow):	not applicable for mixtures	
Vapour pressure:	approx. 5.96 hPa (20 °C) static approx. 17.82 hPa (50 °C) static approx. 20.98 hPa (55 °C) static	
Density:	0.89 g/cm3 (15 °C)	(ISO 2811-3)
	0.879 g/cm3 (20 °C)	(ISO 2811-3)
	0.8581 g/cm3 (50 °C)	(ISO 2811-3)
Relative vapour density (air):	> 1 (20 °C) Heavier than air.	(estimated)

Particle characteristics

Particle size distribution: The substance / product is marketed or used in a non solid or granular form. -

9.2. Other information

Information with regard to physical hazard classes

Explosives

Explosion hazard: not explosive

Oxidizing properties

Fire promoting properties: not fire-propagating

Corrosion to metals

No corrosive effect on metal.

Other safety characteristics

Hygroscopy: Non-hygroscopic
Evaporation rate:
not determined

SECTION 10: Stability and Reactivity

10.1. Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals: No corrosive effect on metal.

10.2. Chemical stability

The product is stable if stored and handled as prescribed/indicated.

Peroxides: 0 %
The product does not contain peroxides.

10.3. Possibility of hazardous reactions

No hazardous reactions when stored and handled according to instructions.

10.4. Conditions to avoid

Avoid all sources of ignition: heat, sparks, open flame. Avoid electro-static discharge.

10.5. Incompatible materials

Substances to avoid:
strong acids, strong bases, strong oxidizing agents

10.6. Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

SECTION 11: Toxicological Information

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Experimental/calculated data:

ATE (oral): > 5,000 mg/kg
ATE (dermal): > 5,000 mg/kg

Information on: Hydrocarbons, C10, aromatics, <1% naphthalene

Experimental/calculated data:
LD50 rat (oral): 6,318 mg/kg (OECD Guideline 401)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: Hydrocarbons, C11-C14, n-alkanes, isoalkanes, cyclics, <2% aromatics

Experimental/calculated data:
LD50 rat (oral): > 5,000 mg/kg (OECD Guideline 420)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: Hydrocarbons, C10, aromatics, <1% naphthalene
Experimental/calculated data:
LC50 rat (by inhalation): > 4.688 mg/l 4 h (OECD Guideline 403)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. The vapour was tested.

Information on: Hydrocarbons, C11-C14, n-alkanes, isoalkanes, cyclics, <2% aromatics
Experimental/calculated data:
LC50 rat (by inhalation): > 5.28 mg/l 4 h (OECD Guideline 403)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. The vapour was tested.

Information on: Hydrocarbons, C10, aromatics, <1% naphthalene
Experimental/calculated data:
LD50 rabbit (dermal): > 2,000 mg/kg (OECD Guideline 402)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: Hydrocarbons, C11-C14, n-alkanes, isoalkanes, cyclics, <2% aromatics
Experimental/calculated data:
LD50 rabbit (dermal): > 2,000 mg/kg (OECD Guideline 402)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Irritation

Assessment of irritating effects:

May cause slight irritation to the skin. May cause slight irritation to the eyes. May cause slight irritation to the respiratory tract.

Respiratory/Skin sensitization

Assessment of sensitization:

Based on available data, the classification criteria are not met.

Information on: Hydrocarbons, C10, aromatics, <1% naphthalene
Assessment of sensitization:
No sensitizing effect.

Information on: Hydrocarbons, C11-C14, n-alkanes, isoalkanes, cyclics, <2% aromatics
Assessment of sensitization:
Skin sensitizing effects were not observed in animal studies. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Germ cell mutagenicity

Assessment of mutagenicity:

Based on the ingredients, there is no suspicion of a mutagenic effect.

Information on: Hydrocarbons, C10, aromatics, <1% naphthalene
Assessment of mutagenicity:
Based on available data, the classification criteria are not met.

Information on: Hydrocarbons, C11-C14, n-alkanes, isoalkanes, cyclics, <2% aromatics
Assessment of mutagenicity:
In the majority of tests performed (bacteria/microorganisms/cell cultures) a mutagenic effect was not found. A mutagenic effect was also not observed in in-vivo assays. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Carcinogenicity

Assessment of carcinogenicity:
Based on available data, the classification criteria are not met.

Information on: Hydrocarbons, C10, aromatics, <1% naphthalene
Assessment of carcinogenicity:
Based on available data, the classification criteria are not met.

Information on: Hydrocarbons, C11-C14, n-alkanes, isoalkanes, cyclics, <2% aromatics
Assessment of carcinogenicity:
Long-term exposure to highly irritating concentrations resulted in skin tumors in animals. A carcinogenic effect in humans can be excluded after brief skin contact. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Reproductive toxicity

Assessment of reproduction toxicity:
Based on the ingredients, there is no suspicion of a toxic effect on reproduction.

Information on: Hydrocarbons, C10, aromatics, <1% naphthalene
Assessment of reproduction toxicity:
Based on available data, the classification criteria are not met.

Information on: Hydrocarbons, C11-C14, n-alkanes, isoalkanes, cyclics, <2% aromatics
Assessment of reproduction toxicity:
The results of animal studies gave no indication of a fertility impairing effect. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Developmental toxicity

Assessment of teratogenicity:

Based on the ingredients, there is no suspicion of a teratogenic effect.

Information on: Hydrocarbons, C10, aromatics, <1% naphthalene
Assessment of teratogenicity:
Based on available data, the classification criteria are not met.

Information on: Hydrocarbons, C11-C14, n-alkanes, isoalkanes, cyclics, <2% aromatics
Assessment of teratogenicity:
No indications of a developmental toxic / teratogenic effect were seen in animal studies. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Possible narcotic effects (drowsiness or dizziness).

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:

Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses.

Aspiration hazard

No aspiration hazard expected.

Interactive effects

No data available.

11.2. Information on other hazards

Endocrine disrupting properties

Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.

Other information

Other relevant toxicity information

The product has not been tested. The statements on toxicology have been derived from the properties of the individual components. Has a degreasing effect on skin.

SECTION 12: Ecological Information

12.1. Toxicity

Assessment of aquatic toxicity:
Harmful to aquatic life with long lasting effects.

Information on:Hydrocarbons, C10, aromatics, <1% naphthalene
Toxicity to fish:
LL50 (96 h) 2 mg/l, Oncorhynchus mykiss

LL50 (96 h) < 5 mg/l, Oncorhynchus mykiss

Information on:Hydrocarbons, C10, aromatics, <1% naphthalene
Aquatic invertebrates:
EL50 (48 h) 3 mg/l, Daphnia magna

EL50 (48 h) < 10 mg/l, Daphnia magna

Information on:Hydrocarbons, C10, aromatics, <1% naphthalene
Chronic toxicity to fish:
NOEL (28 d) 0.441 mg/l, Oncorhynchus mykiss

Information on:Hydrocarbons, C10, aromatics, <1% naphthalene
Chronic toxicity to aquatic invertebrates:
NOEL (21 d) 0.771 mg/l, Daphnia magna

12.2. Persistence and degradability

Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria).

12.3. Bioaccumulative potential

Assessment bioaccumulation potential:
Accumulation in organisms is expected.

Information on:Hydrocarbons, C11-C14, n-alkanes, isoalkanes, cyclics, <2% aromatics
Assessment bioaccumulation potential:
Accumulation in organisms is expected.

12.4. Mobility in soil

Assessment transport between environmental compartments:
Adsorption in soil: No data available.

12.5. Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

12.6. Endocrine disrupting properties

Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.

12.7. Other adverse effects

The product does not contain substances that are listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

Additional information

Other ecotoxicological advice:

The product has not been tested. The statements on ecotoxicology have been derived from the properties of the individual components. Do not discharge product into the environment without control.

SECTION 13: Disposal Considerations

13.1. Waste treatment methods

Must be disposed of or incinerated in accordance with local regulations.

Contaminated packaging:
Uncontaminated packaging can be re-used.Packs that cannot be cleaned should be disposed of in the same manner as the contents.

SECTION 14: Transport Information

Land transport

ADR

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

RID

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Inland waterway transport
ADN

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable

Special precautions for user:	None known

Transport in inland waterway vessel
Not evaluated

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

14.1. UN number or ID number
See corresponding entries for “UN number or ID number” for the respective regulations in the tables above.

14.2. UN proper shipping name
See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

14.3. Transport hazard class(es)
See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.

14.4. Packing group

See corresponding entries for “Packing group” for the respective regulations in the tables above.

14.5. Environmental hazards

See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.

14.6. Special precautions for user

See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.

14.7. Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

SECTION 15: Regulatory Information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

Prohibitions, Restrictions and Authorizations

Annex XVII of Regulation (EC) No 1907/2006: Number on List: 3, 75

Directive 2012/18/EU - Control of Major Accident Hazards involving dangerous substances (EU):
Listed in above regulation: no
Classification applies for standard conditions of temperature and pressure.

15.2. Chemical Safety Assessment

Lead substance(s) exposure scenarios are attached as annex to the safety data sheet.

SECTION 16: Other Information

Assessment of the hazard classes according to UN GHS criteria (most recent version)

Flam. Liq. 4
STOT SE 3 (May cause drowsiness and dizziness.)
Aquatic Acute 3
Aquatic Chronic 3

Full text of the classifications, including the hazard classes and the hazard statements, if mentioned in section 2 or 3:

STOT SE	Specific target organ toxicity — single exposure
Aquatic Chronic	Hazardous to the aquatic environment - chronic
Asp. Tox.	Aspiration hazard
Flam. Liq.	Flammable liquids

Acute Tox.	Acute toxicity
Skin Irrit.	Skin irritation
Eye Irrit.	Eye irritation
Flam. Sol.	Flammable solids
Carc.	Carcinogenicity
Aquatic Acute	Hazardous to the aquatic environment - acute
H336	May cause drowsiness or dizziness.
H412	Harmful to aquatic life with long lasting effects.
H304	May be fatal if swallowed and enters airways.
H411	Toxic to aquatic life with long lasting effects.
H226	Flammable liquid and vapour.
H319	Causes serious eye irritation.
H315	Causes skin irritation.
H332	Harmful if inhaled.
H335	May cause respiratory irritation.
H228	Flammable solid.
H302	Harmful if swallowed.
H351	Suspected of causing cancer.
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.
EUH066	Repeated exposure may cause skin dryness or cracking.

Abbreviations
ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road. ADN = The European Agreement concerning the International Carriage of Dangerous Goods by Inland waterways. ATE = Acute Toxicity Estimates. CAO = Cargo Aircraft Only. CAS = Chemical Abstract Service. CLP = Classification, Labelling and Packaging of substances and mixtures. DIN = German national organization for standardization. DNEL = Derived No Effect Level. EC50 = Effective concentration median for 50% of the population. EC = European Community. EN = European Standards. IARC = International Agency for Research on Cancer. IATA = International Air Transport Association. IBC-Code = Intermediate Bulk Container code. IMDG = International Maritime Dangerous Goods Code. ISO = International Organization for Standardization. STEL = Short-Term Exposure Limit. LC50 = Lethal concentration median for 50% of the population. LD50 = Lethal dose median for 50% of the population. TLV = Threshold Limit Value. MARPOL = The International Convention for the Prevention of Pollution from Ships. NEN = Dutch Norm. NOEC = No Observed Effect Concentration. OEL = Occupational Exposure Limit. OECD = Organization for Economic Cooperation and Development. PBT = Persistent, Bioaccumulative and Toxic. PNEC = Predicted No Effect Level. PPM = Parts per million. RID = The European Agreement concerning the International Carriage of Dangerous Goods by Rail. TWA = Time Weight Average. UN-number = UN number at transport. vPvB = very Persistent and very Bioaccumulative.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Annex: Exposure Scenarios

Index

1. Formulation & (re)packing of substances and mixtures
IS; ERC2; PROC1, PROC2, PROC3, PROC4, PROC5, PROC8a, PROC8b, PROC9, PROC14, PROC15

2. Use in Oilfield drilling and production operations
IS; ERC4; PROC1, PROC2, PROC3, PROC4, PROC8a, PROC8b

3. Use in laboratories
IS; ERC2, ERC4; PROC10, PROC15

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1. Short title of exposure scenario

Formulation & (re)packing of substances and mixtures
IS; ERC2; PROC1, PROC2, PROC3, PROC4, PROC5, PROC8a, PROC8b, PROC9, PROC14, PROC15

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ERC2: Formulation into mixture ESVOC SpERC 2.2.v1: ESVOC SpERC 2.2.v1 PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition PROC4: Chemical production where opportunity for exposure arises PROC5: Mixing or blending in batch processes PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). PROC14: Tabletting, compression, extrusion, pelletisation, granulation PROC15: Use a laboratory reagent.
Operational conditions	
Minimum emission days per year Continuous	100
Emission factor air	1 %

Emission factor water	0.02 %
Emission factor soil	0.01 %
Dilution factor river	10
Dilution factor coast	100
Concentration of the substance	This exposure scenario is based on the exposure scenarios of the above mentioned lead substances.
Risk Management Measures	
Soil treatment measures considered suitable are, e.g.	No application of sludge to soil
	Prevent discharge of undissolved substance to or recover from wastewater
Type of STP	Municipal STP
Estimated subst. removal from wastewater via sewage treatm. (%)	94.6 %
Assumed sewage treatment plant flow (m3/d)	2,000 m3/d
Exposure estimate and reference to its source	
Assessment method	PETRORISK
Maximum amount of safe use	260 t/d

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2. Short title of exposure scenario

Use in Oilfield drilling and production operations
IS; ERC4; PROC1, PROC2, PROC3, PROC4, PROC8a, PROC8b

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article) ESVOC SpERC 4.5a.v1: ESVOC SpERC 4.5a.v1 PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition PROC4: Chemical production where opportunity for exposure arises PROC8a: Transfer of substance or mixture (charging and

	discharging) at non-dedicated facilities PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities
Operational conditions	
Emission factor water	0.0 %
	Discharge to aquatic Environment is restricted by law; the industry prohibits release.
Concentration of the substance	This exposure scenario is based on the exposure scenarios of the above mentioned lead substances.
Risk Management Measures	
	Not applicable as there is no release to wastewater.
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

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3. Short title of exposure scenario

Use in laboratories
IS; ERC2, ERC4; PROC10, PROC15

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ERC2: Formulation into mixture ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article) PROC10: Roller application or brushing PROC15: Use a laboratory reagent.
Operational conditions	
Minimum emission days per year Continuous	20
Emission factor air	2.5 %
Emission factor water	2 %
Emission factor soil	0.01 %
Dilution factor river	10
Dilution factor coast	100
Concentration of the substance	This exposure scenario is based on the exposure scenarios of the above mentioned lead substances.

Risk Management Measures		
Soil treatment measures considered suitable are, e.g.		No application of sludge to soil
	Prevent discharge of undissolved substance to or recover from wastewater	
Type of STP		Municipal STP
Estimated subst. removal from wastewater via sewage treatm. (%)		94.6 %
Assumed sewage treatment plant flow (m3/d)		2,000 m3/d
Exposure estimate and reference to its source		
Assessment method	PETRORISK	
Maximum amount of safe use	2.6 t/d	

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